Kevin Pugliese

(727) 203-1471 | kevingpugliese@gmail.com | linkedin.com/in/kevingpugliese | kevinpugliese.com

EDUCATION

Bachelor of Science in Computer Engineering *December 2027*
University of Florida Honors Program | Gainesville, FL *GPA - 3.67*

Relevant Coursework: *Digital Logic & Computer Systems, Computer Organization, Computational Linear Algebra, Data Structures and Algorithms, Operating Systems*
In Progress: *Digital Design, Microprocessor Applications, Signals & Systems*

EXPERIENCE

Hardware-Software Integration Specialist Co-op *May 2026 - August 2026*
CAE USA, Tampa, Florida

- Replaced an unmaintained 10-year-old C# node inventory tool with a single PowerShell script, cutting a 30-minute manual process required on every shipped simulator to under one minute
- Automated Juniper switch configuration generation by extending an internal Jinja/XML tool; validated output against previously hand-maintained configs
- Traveled to an active Air Force base to support program deployment, performing daily simulator bring-up for customer testing and modifying cockpits to match aircraft hardware changes

Technical Specialist / AI Data Annotator *December 2025 - Present*
Handshake AI Solutions, Remote

- Graded coding agent communication against a written rubric; promoted to reviewer, auditing other annotators' work and flagging LLM-generated submissions
- Designed multistep tasks in KiCad and Defold using proprietary model evaluation tools

PROJECTS

GraphUF (Web App) | *Flask, Git, Nginx, Python, Bash*

- Supplemented UF course catalog with Flask web app to explore transitive course unlocks
- Collaborated with two teammates, handling Git merge conflicts, pull requests, and code reviews
- Processed 8M+ unique data points by scraping and cleaning UF Courses API
- Created transitive closure map of 80k node DAG to achieve O(1) retrieval of reachable nodes

Elementary CPU (Course Project) | *VHDL, Quartus*

- Designed a 4-bit CPU with ROM-based instruction fetch and a registered ALU, achieving 250 MHz Fmax in 57 logic elements and 17 registers on an Intel MAX 10 FPGA
- Implemented the control unit in VHDL as a state machine decoding 6 opcodes into datapath control words, including multi-state instructions for data loads and jumps
- Verified all instructions in simulation and on hardware using hand-assembled test programs

tModLoader (Open Source Contribution) | *C#, GitHub*

- Authored and merged a PR to the official tModLoader repository (10M+ downloads)
- Independently fixed a C# rendering bug that caused some objects to appear misaligned

Battery-Powered Remote LED Switch (Microcontroller Project) | *ESP32, C, Soldering*

- Used 100mAh lithium polymer battery to enable charging and discharging via USB-C
- Optimized hibernation and duty cycles; achieved 12-hour battery life with <1s response time
- Implemented ESP Rainmaker to enable global access to GPIO and manage Wi-Fi connectivity

CERTIFICATIONS

GIAC Foundational Cybersecurity Technologies Certification *August 2023 - August 2027*

- Core Computing Components: Hardware, Virtualization, Networking, Operating Systems, Web, Cloud, Windows and Linux
- Security and Threat Landscape: Exploitation and Mitigation, Forensics and Post Exploitation

SKILLS

Languages: SystemVerilog, VHDL, Assembly, C, C++, Python, C#, Java, PowerShell, Bash
Tools: Git, Linux, Jira, Docker, LTspice, KiCad, Fusion 360, Onshape, Soldering, Quartus, ModelSim